Figure 17.12 shows the algorithm in action: how cost evolves, while cooling the system. While, initially, the cost is fluctuating, the oscillation becomes progressively narrower and, once temperature is low enough, search is channeled toward the global minimum.

We haven't explored all the possible values for α, k, T_0, and even the relative probabilities of applying each of the three transition operators within the ensemble method could be further tuned. To be more systematic, we should build a few more examples with different graphs, and then write a small piece of code that changes one parameter at the time, and records the mean or median cost found: trying it out, using the code included in JsGraph library as a starting point, could be a nice exercise.

17.2.5 Adjacent vs Random Swaps

The last point I'd like to briefly discuss is why it seems that random swapping vertices works better than swapping adjacent pairs.

One factor that contributes to this can be seen with the following example, illustrated in figure 17.13: it shows a case where the local-search transition heuristic, swapping only adjacent pairs, might struggle. First, we have an example of a swap between two random vertices (17.13.A) that grants a good improvement in the cost function: this transition will be always accepted, regardless of the temperature of the system.